PART-1 : PHYSICS

SECTION-I

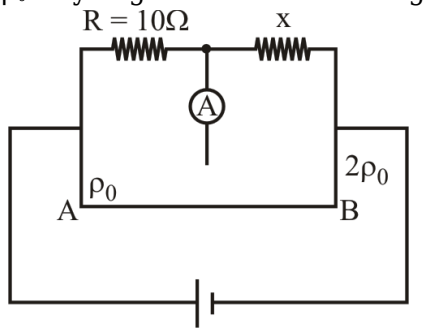
1) A hypothetical experiment conducted to determine Young's modulus formula Y = $\frac{T^x.\tau}{\ell^3}$. If the errors found in □ , τ , T and Y are ±1%, ± 2% , ±1% and 5% respectively then the possible value(s) of x can be:–

(A) x = 0
(B) x = 4
(C) x = 6
(D) x = 10

2) A projectile is thrown with velocity U = 20 m/s ± 5% at an angle 60° to the horizontal. If the projectile falls back on the ground at the same level then which of following can be possible answers for range? Consider g = 10m/s^2.

(A) 39.0 m
(B) 37.5 m
(C) 34.6 m
(D) 32.0 m

3) In a meter bridge experiment, the metre wire has been chosen such that its cross section is uniform but its resistivity increase, linear from ρ_0 to $2\rho_0$ as you go from left end to right end. Neglect any end correction, choose the correct statement(s) :-



(A) If x = 10Ω, Null point is obtained closer to A on wire
(B) If null point is obtained at mid point of AB wire, x will be more than 10Ω
(C) For given value of x null point will not change, even if we change the magnitude of ρ_0
(D) We may get two null points for a particular value of x

4) A conducting spherical shell of inner radius a and outer radius b is made up of a material of resistivity ρ and negligible dielectric activity. At time t = 0, material of shell is electrically neutral and a charge q_0 is some how located at the centre of shell (i.e. at time t = 0, there are no induced charges, they will appear afterward). Choose correct statement(s).

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When $-Q_t$ charge is induced at inner surface at time t, then electric field in the material of shell

(A) is $E = \frac{q_0 - Q_t}{4\pi\epsilon_0 r^2}$ where r is distance of point under consideration from centre of shell.

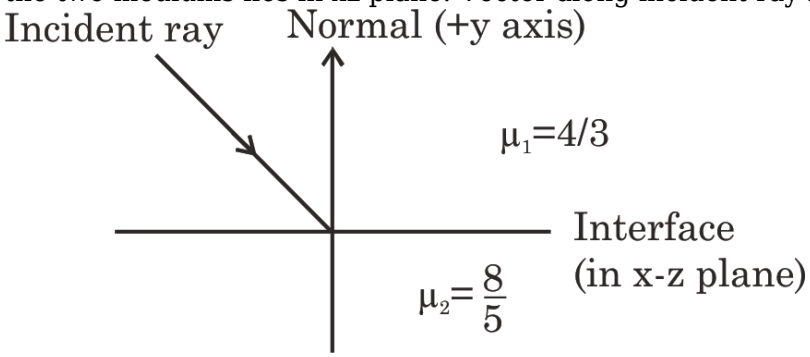
(B) Current through shell at this moment (discussed in option A) is $I = \frac{q_0 - Q_t}{\epsilon_0 \rho}$

At time t we can write

(C) $\frac{dQ_t}{q_0 - Q_t} = \frac{dt}{\epsilon_0 \rho}$

(D) At time t, value of Q_t is given by $Q_t = q_0(1 - e^{-(t/\epsilon_0\rho)})$

5) A light ray incident from a medium of refractive index $\frac{4}{3}$ is shown in diagram. Interface separating the two mediums lies in xz plane. Vector along incident ray is $3\hat{i} - 4\hat{j}$.



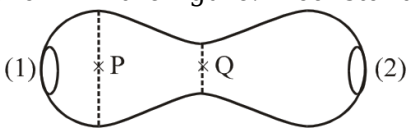
(A) Angle of incidence is 37°.

(B) The angle of refraction is 60º.

(C) The angle of refraction is 30°

(D) If angle of incidence of the incident ray is greater than the critical angle then total internal reflection takes place.

6) A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



(A) The current at the cross-section P equals the current at the cross-section Q

(B) The electric field intensity at P is less than that at Q.

(C) The rate of heat generated per unit time at Q is greater than that at P

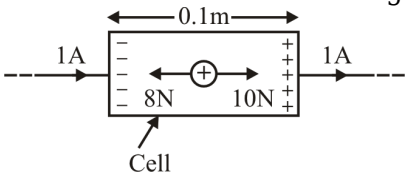
(D) The number of electrons crossing per unit area of cross-section at P is less than that at Q.

SECTION-II

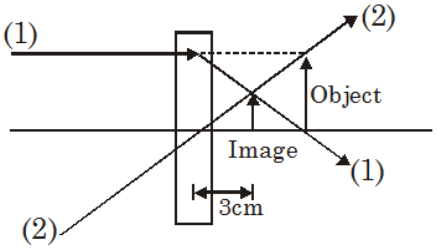
1) An optical bench has 1.5 *m* long scale having four equal divisions in each *cm*. While measuring the focal length of a convex lens, the lens is kept at 75 *cm* mark of the scale and the object pin is kept at 45 *cm* mark. The image of the object pin on the other side of the lens overlaps with image pin that is kept at 135 *cm* mark. In this experiment, the percentage error in the measurement of the focal length of the lens is

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2) Figure shows a cell in which unit positive charge experiences a constant non electric force of 10N and a constant electric force of 8N in directions shown in the figure. The internal resistance of the cell is r Ω. Find the value of 140 r.

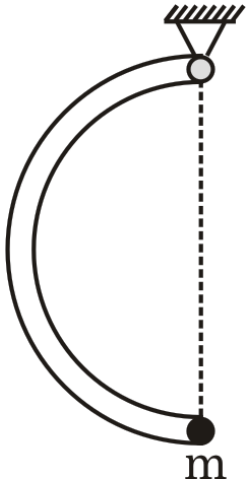


3) The rectangular box is the place of a lens. By looking at the ray diagram calculate the transverse magnification.

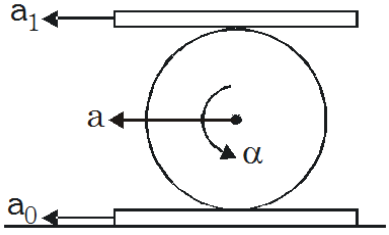


4) A point object is placed 33 cm from a convex mirror of curvature radius = 40 cm. A glass plate of thickness 6 cm and index 2.0 is placed between the object and mirror, close to the mirror. Find the distance (in cm) of final image from the object.

5) Half ring of mass m and radius R is released from the position shown in diagram. A small point mass of same mass is also fixed at the end as shown in figure. If the initial acceleration of point mass m is $\frac{g}{\beta\pi}$ then find the value of β.



6) A system of two planks and a sphere of radius R is in motion as shown in figure. Radius of the sphere is R and there is no slipping anywhere. It is given that $R\alpha = 2a_0$ where α is angular acceleration of sphere and acceleration of upper block is $a_1 = ka_0$ where k is a +ve constant then the value of k will be :-

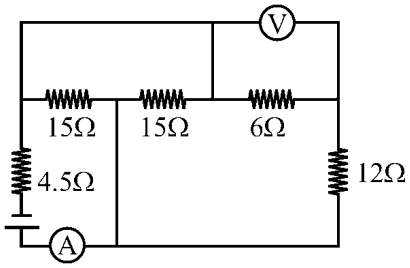


SECTION-III

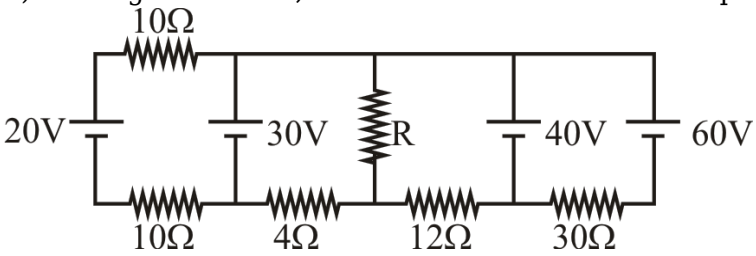
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1) In a vernier callipers, 19 divisions of its main scale match with (20) divisions on its vernier scale. Each division of the main scale is '*a*' units. Using the vernier principle, if least count is $\frac{a}{4x}$ then find the value of x.

2) A galvanometer of coil resistance 1Ω is converted into voltmeter by using a resistance of 5Ω in series and another identical galvanometer is converted into ammeter by using a shunt of 1Ω. Now ammeter and voltmeter connected in circuit as shown, find the ratio of reading of voltmeter (in volts) and ammeter (in amperes).



3) In the given circuit, the value of R so that thermal power generated in R will be maximum (in Ω).



4) A thin rod of mass m and length 2□ is placed horizontally and perpendicular on a horizontal peg, and is set free to move in vertical plane. The point of contact of the rod is □/3 distance away from the centre of rod. If the rod starts slipping when it forms an angle θ with the horizontal then if the coefficient of friction of rod with peg is N tanθ, then find the value of N.

5) A uniform 8 kg door with dimensions 75cm × 200cm is pivoted at hinges 25 cm from the top and from the bottom. The horizontal force exerted by each hinge is 10αN. The value of α is (Assume centre of mass of gate at its geometrical centre)



6) An equi-convex lens of focal length 10cm in air and refractive index (μ_g = 1.5), is placed in a liquid whose refractive index varies with time as $\mu(t) = \left[1.0 + \frac{1}{10}t\right]$, (t in sec). Lens will act as diverging lens of focal length 20 cm at t = 5n (in sec). Find n.

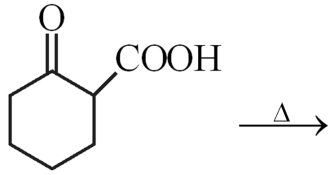
PART-2 : CHEMISTRY

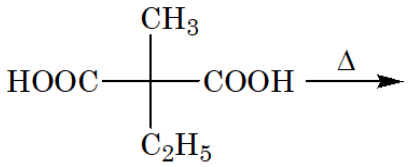
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SECTION-I

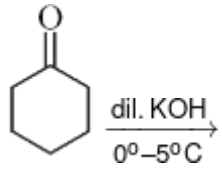
1)

Which of the following will give product having one chiral atom after reaction is completed -

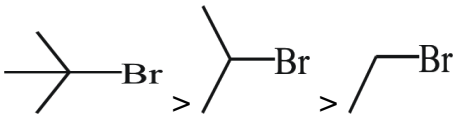
(A) 

(B) 

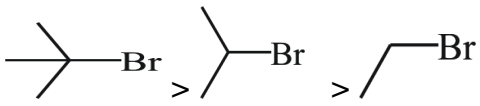
(C) $CH_3-CHO \xrightarrow{HCN}$

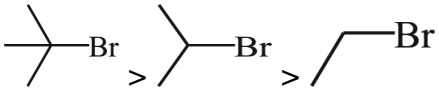
(D) 

2) The correct statement is-

(A) Order of reactivity of E_1 is as : 

(B) Order of nucleophilicity $F^- < Cl^- < Br^- < I^-$ (In H_2O)

(C) Order of reactivity of S_N2 is as : 

(D) Order of reactivity of S_N1 is as : 

3) Which one of following pairs of aqueous solution will be isotonic at same temperature

(A) 1M NaCl and 2 M urea

(B) 1.5 M $AlCl_3$ and 2M Na_2SO_4

(C) 2.5 M KCl and 1M $Al_2(SO_4)_3$

(D) 1M $CaCl_2$ and 1M NaCl

4) The equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$ is attained at 25°C in a closed container and inert gas helium is introduced at constant volume. Which of the following statements is/are not correct :-

(A) Concentrations of SO_2, Cl_2 and SO_2Cl_2 change

(B) More chlorine is formed

(C) Concentration of SO_2 is reduced

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(D) More SO_2Cl_2 is formed

5) A greenhouse effect causing gas which turns lime water milky, when is treated with carbon gives compound B. Which of the following statements is correct regarding B?

(A) It binds with haemoglobin

(B) It is also green house effect causing gas

(C) It is one of serious air pollutant

(D) It is lethal gas

6) Which of the following is/are correct about D_2O.

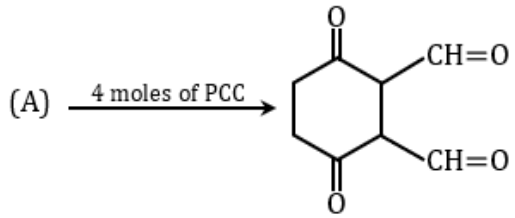
(A) 100 ml of its weights is more than 100 gm

(B) Acts as a moderator

(C) Boiling point of D_2O is 373 K

(D) It is used in exchange reactions for the study of reaction mechanism

SECTION-II



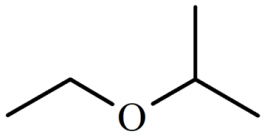
1)
Maximum number of moles of Ac_2O consumed by reactant (A) is :

2) How many ether will give two alkyl iodides on reaction with excess amount of HI.

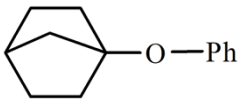
(i) MeOEt

(ii) Ph-OMe

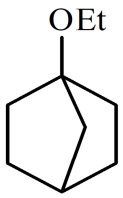
(iii)



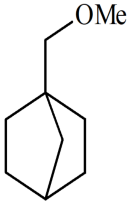
(iv)



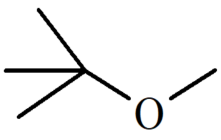
(v)



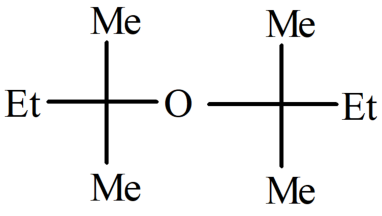
(vi)



(vii)

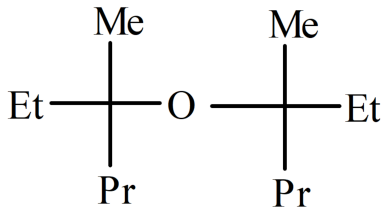


(viii)



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(ix)



3) 20 mL of 0.1 M NaOH is added to 50 mL of 0.1 M acetic acid solution. The pH of the resulting solution is _____ $\times 10^{-2}$ (Nearest integer)
Given : pKa (CH_3 COOH) = 4.76
log 2 = 0.30
log 3 = 0.48

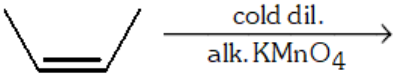
4) A complex is $CoCl_3.xNH_3$. Its 0.1m solution in water shows $\Delta T_f = 0.558°C$. If complex is 100% ionised and coordination number is 6, find the value of 'x'. ($K_{f(H_2O)}$ = 1.86 K Kg/mole)

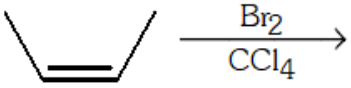
5) Find the number of pair(s) in which one of the ion is possible to be separated from other by using dil.HCl + H_2S.
(i) Bi^{3+} , Al^{3+} (ii) Al^{3+}, Cr^{3+} (iii) Cd^{2+} , Cu^{2+} (iv) Zn^{2+} , Cu^{2+} (v) Zn^{2+}, Mn^{2+} (vi) Mg^{2+}, Pb^{2+}

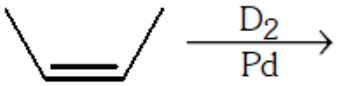
6) $CuSO_4$ decolourises on addition of excess KCN, in the product :
a = atomicity of the product
b = number of atomic orbitals involved in hybridisation of copper in product
c = number of 3d electrons in copper in the product
Find the value of a + b + c

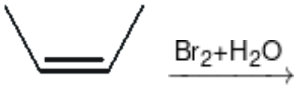
SECTION-III

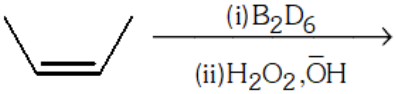
1) Consider the following reactions.

(i) 

(ii) 

(iii) 

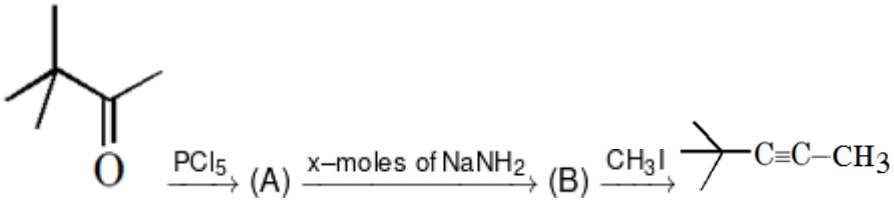
(iv) 

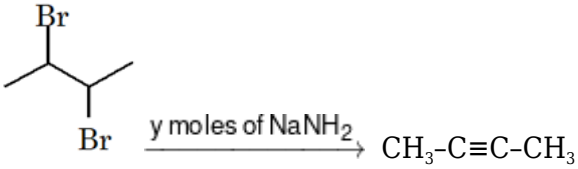
(v) 

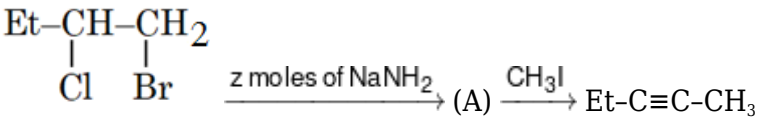
(vi) CH_3–C≡C–CH_3 $\xrightarrow[liq.NH_3]{Na}$

Find number of reactions which give final major product selectively through anti addition.

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2) **Reaction-I :** 

Reaction-II : 

Reaction-III : 

Value of x + y + z is

3) An acidic aqueous solution contains 0.02 M – $FeCl_2$ and 0.05 M – $FeCl_3$, at 25°C. Now NaOH solution is added is this solution , drop by drop. The pH range in which only the precipitation of $Fe(OH)_3(s)$ occur with out precipitation of $Fe(OH)_2(s)$ differs by 'x' unit . The value of 'x' is given :

K_{sp} of $Fe(OH)_2 = 8 \times 10^{-16}$

K_{sp} of $Fe(OH)_3 = 4 \times 10^{-28}$

4) Two solid compounds A and C dissociate into gaseous product at temperature T as follows:

(i) $A(s) \rightleftharpoons B(g) + D(g)$; $Kp_1 = 625\ (atm)^2$

(ii) $C(s) \rightleftharpoons E(g) + D(g)$; $Kp_2 = 975\ (atm)^2$

When both solids are present in same container, in large excess, then total pressure (in atm) over the solid mixture is P. The value of $\frac{P}{10}$ is :

5) How many of the following acidic radical(s) give(s) coloured gas with dil. H_2SO_4 ?

NO_3^- ; Cl^- ; Br^- ; I^- ; NO_2^- ; CO_3^{2-} ; SO_3^{2-} ; CH_3COO^-

6) Find the number of compound(s) which are green coloured .

MnO_2, K_2MnO_4, Cr_2O_3, $Cu(SO_4).5H_2O$, CuO, $CuCO_3.Cu(OH)_2$, $FeSO_4.7H_2O$

PART-3 : MATHEMATICS

SECTION-I

1) If $f(x) = \lim_{n\to\infty}\left(\frac{\cos \pi x - x^{2n}\sin(x-1)}{1 + x^{2n+1} - x^{2n}}\right)$, $n \in N$, then f(x) is (where [.] represent greatest integer function)

(A) continuous at x = 0

(B) continuous at x = –1

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(C) $\left[\lim_{x\to 1} f(x)\right] = -1$

(D) $\left[\lim_{x\to 1^+} f(x)\right] = 1$

2) Let $f(x) = \max\{1 + \sin x, 1, 1 - \cos x\}\ \forall\ x \in [0, 2\pi]$ and $g(x) = \max\{1, |x - 1|\}\ \forall\ x \in R$. Then

(A) $g(f(0)) = 1$

(B) $g(f(1)) = 1$

(C) $f(g(1)) = 1 + \sin 1$

(D) $f(g(x))$ is differentiable at $x = 0$

3) Let $f : A \to B$ be a function where $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4\}$. Then identify the correct statements.

(A) No. of onto function with $f(1) = 1$ and $f(2) = 2$ are 18.

(B) No. of function with $f(i) \neq 1\ \forall\ i \in B$ are 324

(C) No. of function satisfying $f(1) \le f(2) \le f(3) \ge f(4) \ge f(5)$ are 55

(D) No. of function satisfying $f(1) \le f(2) \le f(3) \ge f(4) \ge f(5)$ are 371

4) Let $f(x) = \begin{cases} \min\left(\left\{x, x^2\right\}\right) & , \quad x \geqslant 0 \\ \max\left\{2x, x^2 - 1\right\} & , \quad x < 0 \end{cases}$ then which of the following is/are true ?

(A) $f(x)$ is continuous at $x = 0$

(B) $f(x)$ is not differentiable at $x = 1$

(C) $f(x)$ is not differentiable at exactly three point

(D) Range of $f(x)$ is $[-1, \infty)$

5) Let $f : N \to N$, where $f(x) = x + (-1)^{x-1}$. Identify the correct statement(s).

(A) $f(x)$ is many one function and hence non invertible

(B) $f(x)$ is one one function but not onto

(C) $f(f(f(f(x)))) = x\ \forall\ x \in N$

(D) Graphs of $f(x)$ and $f^{-1}(x)$ meet at infinitely many points

6) Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular unit vectors and a unit vector $\vec{r}$ satisfying the equation $\left(\vec{b}-\vec{c}\right) \times (\vec{r} \times \vec{a}) + (\vec{c}-\vec{a}) \times \left(\vec{r} \times \vec{b}\right) + \left(\vec{a}-\vec{b}\right) \times (\vec{r} \times \vec{c}) = \vec{0}$ can be equal to

(A) $\frac{1}{\sqrt{3}}\left(\vec{a}+\vec{b}+\vec{c}\right)$

(B) $-\frac{1}{\sqrt{14}}\left(2\vec{a}+3\vec{b}+\vec{c}\right)$

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(C) $\frac{1}{\sqrt{14}}\left(2\vec{a}+3\vec{b}+\vec{c}\right)$

(D) $-\frac{1}{\sqrt{3}}\left(\vec{a}+\vec{b}+\vec{c}\right)$

SECTION-II

1) The sum of all positive integral values of 'a', $a \in [1, 500]$ for which the equation $[x]^3 + x - a = 0$ has real solution is (where [.] denote greatest integer function)

2) Let $\lim_{x\to\infty} x^a\left(\sqrt[3]{x+1}+\sqrt[3]{x-1}-2\sqrt[3]{x}\right)=\lambda$ where λ is some non-zero real number. Then the value of $(a + \lambda)$ can be expressed as p/q where p and q are relatively prime positive integers. Then $(p + q)$ equals

3) Let the base of a pyramid is rectangular, three of its vertices of base are A(2, 2, –1), B(3, 1, 2) and C(1, 1, 1). Its vertex at the top is $P\left(4, \frac{-26}{3}, \frac{-10}{3}\right)$. Let the coordinates of foot of normal drawn from P on the base of pyramid is (ℓ, m, n) then $\frac{\ell+m+n}{10}$ is

4) Find the value of $\frac{\sum_{r=1}^{\infty}\tan^{-1}\left(\frac{6.7^{(r-1)}}{1+7^{(2r-1)}}\right)}{\pi}$

5) Let $f(x)=\begin{cases} x^3-6x & x\in Q \\ a-3x & x\in \bar{Q}\end{cases}$

(Q : rational, $\bar{Q}$: irrational number)

If f(x) is continuous at three point, then number of integral values of a

6) If $f : R^+ \to R^+$ a function such that $f(x)\,f(y) = f(xy) + 2\left(x+\frac{1}{y}+1\right)$, then $\sum_{r=1}^{5}\left(f(r^2)-f(r)\right)$ =

SECTION-III

1) The number of solutions of $\tan^{-1}(x+2)+\cot^{-1}\sqrt{4x+20}=\lim_{x\to\frac{\pi}{2}}\frac{\sin(x\cos x)}{\cos(x\sin x)}$ in [–5, 5] is

2) A differentiable function f(x) satisfies $f(xy) = (f(x))^y \ \forall\ x, y \in R$ and $f(x) = \frac{1}{4}$. If

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$\lim_{n\to\infty}\left(1.\sqrt{f(x+1)}+2.\sqrt{f(x+2)}+...+n\sqrt{f(x+n)}\right)=k\sqrt{f(x)}$, then k =

3) Maximum value of the function $f(x)=\frac{2x^4-14x^2-8x+49}{x^4-7x^2-4x+23}$ is

4) $f(x) = \{x + [\sin x]\} + \{x + [\sin^2 x]\} + \{x + [\sin^3 x]\} + ... + \{x + [\sin^{10} x]\}$, then [f(e)] equals (where [.] & {.} denotes greatest integer & fractional part function respectively)

5) There is a unique differentiable function f from R to R satisfying $f(x) + (f(x))^3 = x + x^7$ for all real x. The derivative of f(x) at x = 2 can be expressed as a common fraction a/b. Where a and b are relatively prime positive integers. Then $\left(\frac{a+b}{75}\right)$ equals

6) Let A_1, A_2, A_3, A_4 be the areas of the triangular faces of a tetrahedron (not regular) and h_1, h_2, h_3, h_4 be the corresponding altitudes of the tetrahedron. If volume of the tetrahedron is 5 cubic units, then $\frac{1}{60}[(A_1 + A_2 + A_3 + A_4)](h_1 + h_2 + h_3 + h_4))$ (in cubic units) has min. value equal to

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ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6
A.	A,B,C,D	B,C,D	B,C	A,B,C,D	A,C	A,B,C,D

SECTION-II

Q.	7	8	9	10	11	12
A.	1.35 to 1.45	28.00	0.50	42.00	1.50	5.00

SECTION-III

Q.	13	14	15	16	17	18
A.	5	1	3	2	2	2

PART-2 : CHEMISTRY

SECTION-I

Q.	19	20	21	22	23	24
A.	B,C,D	A,B,D	A,B,C	A,B,C,D	A,C,D	A,B,D

SECTION-II

Q.	25	26	27	28	29	30
A.	4.00	5.00	458.00	5.00	3.00	22.00

SECTION-III

Q.	31	32	33	34	35	36
A.	3	8	2	8	1	4

PART-3 : MATHEMATICS

SECTION-I

Q.	37	38	39	40	41	42
A.	A,C	A,B,C	A,B,D	A,B,C	C,D	A,D

SECTION-II

Q.	43	44	45	46	47	48
A.	812.00	22.00	0.40	0.25	3.00	40.00

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SECTION-III

Q.	49	50	51	52	53	54
A.	**1**	**2**	**3**	**7**	**7**	**4**

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SOLUTIONS

PART-1 : PHYSICS

1) $= \pm\frac{\Delta Y}{Y} \pm x\left(\frac{\Delta T}{T}\right) \pm \frac{3\Delta T}{\tau} \pm \frac{\Delta \ell}{\ell} \Rightarrow \pm 5 = \pm x \pm 2 \pm 3$

$\Rightarrow x = 0, 4, 6, 10$

2) $R = \frac{20^2 \times \sin 120^\circ}{g} = 20\sqrt{3} = \frac{\Delta R}{R} = \frac{24U}{U}$

$\Rightarrow \Delta R = \frac{2 \times 5}{100} \times 200\sqrt{3} = 2\sqrt{3}$

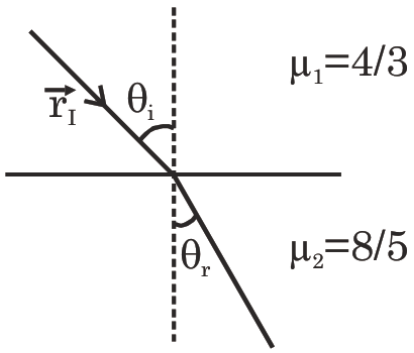
$20\sqrt{3} - 2\sqrt{3} < R < 20\sqrt{3} + 2\sqrt{3} \Rightarrow 31.1\text{m} < R < 38.1\ \text{m}$

4) $E\,4\pi r^2 = \frac{q_0 - Q_t}{\in_0}$

$J = \frac{E}{\rho} = \frac{q_0 - Q_t}{4\pi \in_0 \rho r^2}$

$I = JA = \frac{1}{\in_0 \rho}(q^0 - Q)$

$I = \frac{dQ}{dt} \Rightarrow \frac{dQ}{q_0 - Q} = \frac{dt}{\in_0 \rho} \Rightarrow \frac{Q}{q_0} = 1 - e^{-t/\varepsilon_0 \rho}$



5)

$\vec{r}_I = +3\hat{i} - 4\hat{j}$

$\tan\theta_i = \frac{3}{4}$

$\theta_i = 37°$

$\mu_1 \sin\theta_i = \mu_2 \sin\theta_r$

$\frac{4}{3} \times \frac{3}{5} = \frac{8}{5}\sin\theta_r$

$\sin\theta_r = \frac{1}{2}$

$\theta_r = 30°$

7)

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$1MSD = \frac{1}{4}cm \Rightarrow \Delta v = \Delta u = 2\left(\frac{1}{4}cm\right) = \frac{1}{2}cm$

$X_{lens} = 75cm, X_{object} = 45cm, X_{image} = 135cm$

$u = -(75-45) = -30cm and v = (135-75) = 60cm$

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{60} + \frac{1}{30} \Rightarrow f = 20cm$

Also $\frac{\Delta f}{f^2} = \frac{\Delta v}{v^2} + \frac{\Delta u}{u^2} \Rightarrow \frac{\Delta f}{f}\% = \left(\frac{\Delta v}{v^2} + \frac{\Delta u}{u^2}\right) \times f \times 100\%$

$\frac{\Delta f}{f}\% = \left(\frac{1/2}{(60)^2} + \frac{1/2}{(30)^2}\right) 20 \times 100\% \approx 1.39\%$

8) V = Potential difference across the cell = Electric field x width of the cell
= 8 × 0.1 = 0.8 volt
ε = emf of the cell = 10 × 0.1 = 1.0 volt
Also r is the internal resistance and i is the current drawn from the cell
V = ε – ir or 0.8 = 1 – 1r
⇒ r = 0.2 Ω

9)

From ray diagram object is placed at focus of lens and lens is concave.

$\frac{1}{\nu_1} - \frac{1}{u} = \frac{1}{f}$

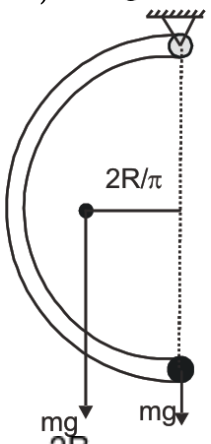
$\frac{1}{(-3)} - \frac{1}{(-f)} = \frac{-1}{f}$

$\frac{-1}{3} = \frac{-2}{f}$

f = 6 cm

$m = \frac{v}{u} = \frac{3}{6} = \frac{1}{2}$

11) $\tau = I_0\alpha$

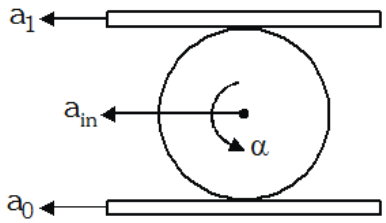


$mg\frac{2R}{\pi} = \left(2mR^2 + 4mR^2\right)\alpha$

$\alpha = \frac{mg2R}{\pi 6mR^2} = \frac{g}{3\pi R}; a_m = 2R \times \alpha = \frac{2g}{3\pi}$

12) $a_0 = a_{cm} - R\alpha$ ....(1)

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$a_1 = a_{cm} + R\alpha$ ....(2)

Solving eqn. (1) & (2)

$a_1 - a_0 = 2R\alpha = 4a_0$

$a_1 = 5a_0$

K = 5.00

13) 1 *MSD* = *a*

19 *MSD* = 20 *VSD*

$$\Rightarrow 1VSD = \frac{19}{20} MSD$$

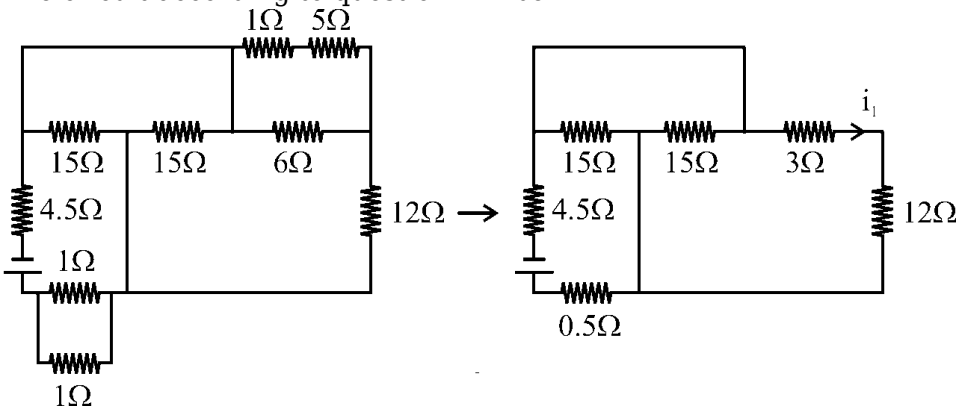
$$\Rightarrow 1VSD = \frac{19}{20} a$$

$$\Rightarrow L.C. = a - \frac{19a}{20} = \frac{a}{20}$$

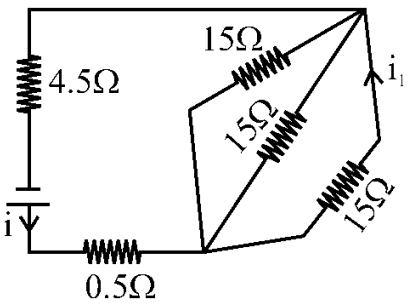
$$\Rightarrow L.C. = \frac{a}{20} = \frac{a}{4x} \Rightarrow x = 5$$

14)

The circuit according to question will be



It can be redrawn as



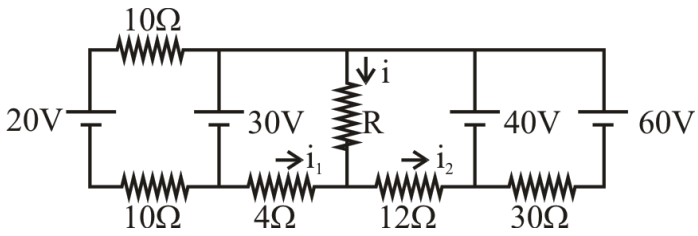
Let reading of ammeter be i. Then i_1 = i/3 clearly.

Reading of voltmeter is V = i_1 × 3Ω = i/3 × 3 = i

V/i = 1.

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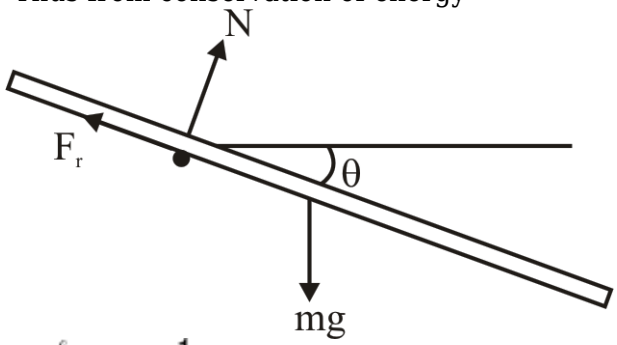
15)



$R_{eq} = R_{in}$

$$= \frac{4 \times 12}{4 + 12} = 3\Omega$$

16) Till the rod doesn't slip the centre of mass of the rod is in circular motion around the peg.
Thus from conservation of energy



$$mg\frac{\ell}{3}\sin\theta = \frac{1}{2}I\omega^2$$

Where $I = \frac{4m\ell^2}{12} + \frac{m\ell^2}{9} = \frac{4m\ell^2}{9}$

thus $\omega = \sqrt{\frac{3g}{2\ell}\sin\theta}$

Now $m\frac{\ell}{3}\omega^2 = F_r - mg\sin\theta$

or $F_r = \frac{3}{2}mg\sin\theta$

Angular acceleration of rod at an angle θ

$$\frac{mg\frac{\ell}{3}\cos\theta}{\frac{4m\ell^2}{9}} = \alpha$$

$$\Rightarrow \alpha = \frac{3}{4}\frac{g}{\ell}\cos\theta$$

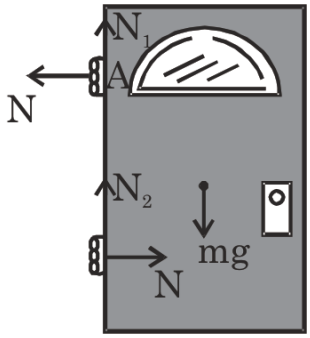
and linear acceleration of centre of mass perpendicular to rod ⇒ mg cos θ – N = ma
From pure rolling a = □/3 α thus

$$a = \frac{g}{4}\cos\theta$$

$$\Rightarrow N = \frac{3mg}{4}\cos\theta$$

$$\mu = \frac{F_r}{N} = 2\tan\theta$$

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17)

Balancing torque w.r.t point A

$$N.150 = mg \times \frac{75}{2}$$

$$N = \frac{mg}{4} = \frac{8 \times 10}{4} = 20$$

18) $$\frac{1}{10} = \left(\frac{1.5}{1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)\frac{1}{-20} = \left(\frac{1.5}{\mu} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$-2 = \frac{1.5 - 1}{\left(\frac{1.5}{\mu} - 1\right)}$$

$$-2\left(\frac{1.5}{\mu} - 1\right) = 1.5 - 1$$

$$\frac{-3}{\mu} + 2 = 0.5$$

$$\frac{-3}{\mu} = -1.5$$

$$\mu = 2 = 1 + \frac{1}{10}t$$

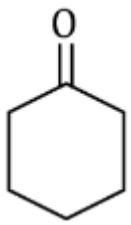
$$\frac{1}{10}t = 1$$

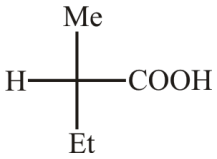
$$t = 10 = 5n$$

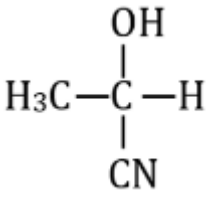
$$n = 2 \text{ sec}$$

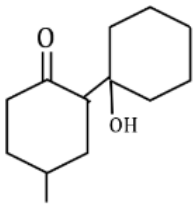
PART-2 : CHEMISTRY

19)

A : 

B : 

C : 

D : 

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20)

Sn_1, E_1 reactivity ∝ stability of carbocation

21)

$\pi = i \times c \times R \times T$

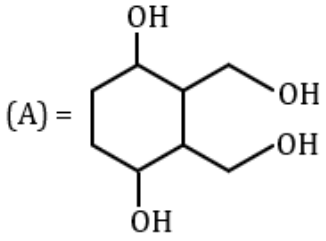
(A) $\pi_1 = 2RT$ ——— $\pi_2 = 2RT$ (isotonic)

(B) $\pi_1 = 6RT$ ——— $\pi_2 = 6RT$ (isotonic)

(C) $\pi_1 = 5RT$ ——— $\pi_2 = 5RT$ (isotonic)

(D) $\pi_1 = 3RT$ ——— $\pi_2 = 2RT$ (Not an isotonic)

24) Theory based.

25) (A) = 

No. of alcohol = No. of moles of Ac_2O consumed = 4

26) i, iii, vii, viii, ix, will give two iodides.

27)

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

Initially	5mmol	2mmol	0	0
after Rxn	3mmol	0	2 mmole	2 mmole

$$pH = pKa + \log_{10}\frac{[salt]}{[acid]}$$

$$pH = 4.76 + \log_{10}\frac{2}{3}$$

$pH = 4.58 = 458 \times 10^{-2}$

28) $\Delta T_f = k_f \times m \times i$

$i = 3$

∴ Complex must be $[COCl.xNH_3].Cl$

∴ $x = 5$

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30) a = 8
b = 4
c = 10
$K_3[Cu(CN)_4]$
+1
$Cu^+ = [Ar]3d^{10}4s^0$
Hybridisation state of $Cu^+ = sp^3$

31) ii, iv, vi

32)

x = 3, y = 2, z = 3
x + y + z = 8

33) $[OH^-]_{min}$ to ppt $Fe(OH)_2 = \left(\frac{8 \times 10^{-16}}{0.02}\right)^{\frac{1}{2}} = 2 \times 10^{-7}$
$\Rightarrow P^H_{min} = 7.3$
$[OH^-]_{min}$ to ppt $Fe(OH)_3 = \left(\frac{4 \times 10^{-28}}{0.05}\right)^{\frac{1}{3}} = 2 \times 10^{-9}$
$\Rightarrow P^H_{min} = 5.3$
$\therefore \Delta pH = 7.3 - 5.3 = 2.0$ unit

34) $P_T = 2\sqrt{K_{P1} + K_{P2}}$

35) $NO_2^- \xrightarrow{\text{dil } H_2SO_4} \underset{\text{Brown gas}}{NO_2 \uparrow}$

36) Green colour compounds are K_2MnO_4 , Cr_2O_3, $Cu(OH)_2$, $CuCO_3$, $FeSO_4.7H_2O$

PART-3 : MATHEMATICS

37) $f(x) = \lim_{n\to\infty}\left(\frac{\cos \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}\right)$

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$$= \begin{cases} \frac{\sin(x-1)}{1-x} & , \quad x < -1 \\ 1-\sin 2 & , \quad x = -1 \\ \cos \pi x & , \quad -1 < x < 1 \\ -1 & , \quad x = 1 \\ \frac{\sin(x-1)}{1-x} & , \quad x > \end{cases}$$

$= \lim_{x \to 1^-} f(x) = -\frac{\sin 2}{2}$

$= \lim_{x \to 1^+} f(x) = -1$

 f(x) is not continuous at x = –1

f(x) is continuous at x = 0

38)

$$f(x) = \begin{cases} 1+\sin x & , \quad 0 \leqslant x \leqslant 3\pi/4 \\ 1-\cos x & , \quad 3\pi/4 \leqslant x \leqslant 3\pi/2 \\ 1 & , \quad 3\pi/2 \leqslant x \leqslant 2\pi \end{cases}$$

$$g(x) = \begin{cases} 1-x & , \quad x \leqslant 0 \\ 1 & , \quad 0 \leqslant x \leqslant 2 \\ x-1 & , \quad x \geqslant 2 \end{cases}$$

f(0) = 1 ⇒ g(f(0)) = 1

f(1) = 1 + sin1 $0 < 1 < \frac{3\pi}{4}$

⇒ g(f(1)) = 1 1 < 1 + sin1 < 2

again g(1) = 1 ⇒ f(g(1)) = 1 + sin1

g(0) = 1 ⇒ f(g(0)) = 1 + sin1

39)

(A) No. of function = Total [when atleast one of 3,4 not used]

$= 4^3 - [{}^2C_1.3^2 - {}^2C_2 2^3]$

= 64 – [54 – 8] = 18

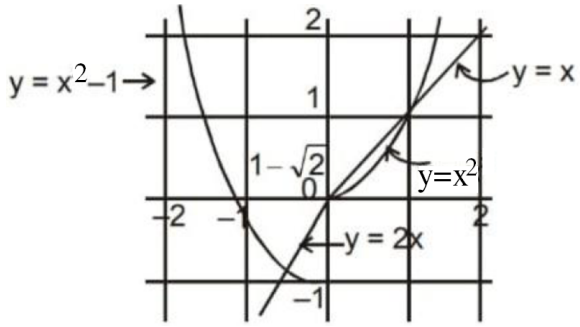
(B) 4×3^4

(C) If

f(3)	Total function
1	1
2	9
3	36
4	100
5	225

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40)



41) $f(x) = \begin{cases} x-1 & x = \text{even} \\ x+1 & x = \text{odd} \end{cases}$

$f^{-1}(x) = \begin{cases} x+1 & x = \text{Odd} \\ x-1 & x = \text{Even} \end{cases}$

42) $\left(\vec{b}-\vec{c}\right) \times (\vec{r} \times \vec{a}) = \left(\left(\vec{b}-\vec{c}\right).\vec{a}\right)\vec{r} - \left(\left(\vec{b}-\vec{c}\right).\vec{r}\right)\vec{a}$

Now, $\left(\vec{b}-\vec{c}\right) \times (\vec{r} \times \vec{a}) + (\vec{c}-\vec{a}) \times \left(\vec{r} \times \vec{b}\right) + \left(\vec{a}-\vec{b}\right) \times (\vec{r} \times \vec{c}) = 0$

$\Rightarrow \left[\left(\vec{b}-\vec{c}\right).\vec{a} + (\vec{c}-\vec{a}).\vec{b} + \left(\vec{a}-\vec{b}\right).\vec{c}\right]\vec{r} - \left[\left(\vec{b}-\vec{c}\right).\vec{r}\right]\vec{a} - [(\vec{c}-\vec{a}).\vec{r}]\,\vec{b} - \left[\left(\vec{a}-\vec{b}\right).\vec{r}\right]\vec{c} = 0$

$\Rightarrow \left(\vec{b}.\vec{r}-\vec{c}.\vec{r}\right)\vec{a} + (\vec{c}.\vec{r}-\vec{a}.\vec{r})\,\vec{b} + \left(\vec{a}.\vec{r}-\vec{b}.\vec{r}\right)\vec{c} = 0$

$\Rightarrow \vec{a}.\vec{r} = \vec{b}.\vec{r} = \vec{c}.\vec{r}$

43) Let $x = n$

$a = n + n^3$

$\forall\, n \in \{1, 2, 3, \ldots, 7\}$

$\sum a = \sum n + \sum n^3$

$= \frac{7(7+1)}{2} + \left(\frac{7(7+1)}{2}\right)^2$

$= 812$

44)

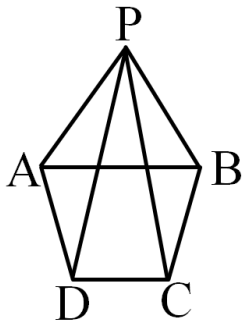
Let $x = \frac{1}{t} \quad t \to 0^+$

$\lambda = \lim\limits_{t\to 0^+} \frac{(1+t)^{1/3} + (1-t)^{1/3} - 2}{t^{a+\frac{1}{3}}}$

$= \lim\limits_{t\to 0^+} \frac{\left(1 + \frac{1}{3}t + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2}t^2\right) + \left(1 - \frac{1}{3}t + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2}t^2\right) - 2}{t^{a+\frac{1}{3}}}$

$\Rightarrow \lambda = -\frac{2}{9} \quad a = \frac{5}{3}$

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45)

Equation of ABCD plane = x – 5y – 2z + 6 = 0

For foot of perpendicular

$$\Rightarrow \frac{x-4}{1} = \frac{y+\frac{26}{3}}{-5} = \frac{z+\frac{10}{3}}{-2} = -\left(\frac{60}{30}\right)$$

$$\ell + m + n = 4 - \frac{26}{3} - \frac{10}{3} - 2(-6) = 4$$

46) $\tan^{-1}\left(\frac{6.7^{r-1}}{1+7^{2r-1}}\right)$

$= \tan^{-1}7^r - \tan^{-1}7^{r-1}$

sum $= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

47)

$x^3 - 6x = a - 3x$

$x^3 - 3x = a$

has 3-distinct real roots

48)

Put x = 1, y = 1

$(f(1))^2 = f(1) + 6 \Rightarrow f(1) = 3, -2$

f(1) = 3 [Since f(x) > 0]

Put y = 1 in given relation

f(x) f(1) = f(x) + 2(x + 2)

2f(x) = 2x + 4

f(x) = x + 2

$$\sum_{r=1}^{5}\left(f\left(r^2\right) - f(r)\right) = \sum_{r=1}^{5}\left(r^2 - r\right) = 40$$

49)

$$\lim_{x\to\frac{\pi}{2}} \frac{\sin(x\cos x)}{\cos((x\sin x))} \left[\frac{0}{0}\text{form}\right]$$

$$\lim_{x\to\frac{\pi}{2}} \frac{\cos(x\cos x).(\cos x - x\sin x)}{\sin(x\sin x).(\sin x + x\cos x)} = \frac{\pi}{2}$$ [Using L'Hospital Rule]

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Given equation becomes

$$\tan^{-1}(x+2)+\cot^{-1}\left(\sqrt{4x+20}\right)=\frac{\pi}{2}$$

$\Rightarrow x+2=\sqrt{4x+20}$

$\Rightarrow x=4$

50)

$f(xy)=(f(x))^y$

Put $x=1$

$f(y)=(f(1))^y$

$$f(y)=\frac{1}{4^y}$$

$$L=\lim_{n\to\infty}\frac{1}{2^{x+1}}+\frac{1}{2^{x+2}}+\ldots+\frac{n}{2^{x+n}}$$

$$=\frac{1}{2^x}\left[\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\ldots\infty\right]$$

$$k=\frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\ldots.\infty$$

$$k=\frac{1}{2^2}+\frac{2}{2^3}+\ldots.\infty$$

$$\frac{k}{2}=\frac{1}{2}+\frac{1}{2^2}+\ldots.=\frac{\frac{1}{2}}{1-\frac{1}{2}}$$

$\Rightarrow k=2$

51) $$f(x)=2+\frac{3}{x^4-7x^2-4x+23}$$

Let $h(x)=x^4-7x^2-4x+23$

$=(x^2-4)^2+(x-2)^2+3$

$\Rightarrow$ Range of $h(x)$ $[3,\infty)$

$\Rightarrow$ Range of 'f' $(2,3]$

52)

$f(x)=10\{x\}$

$f(e)=10\{e\}=10\{2.71\}=7.1$

$[f(e)]=7$

53)

We use implicit differentiation. The equation $f(x)+(f(x))^3=x+x^7$ holds for all $x\in R$, so both sides of this equation have the same derivative with respect to x. By the Chain Rule, this means

$f'(x)+3(f(x))^2.f'(x)=1+7x^6\Rightarrow(1+3(f(x))^2)\,f'(x)=1+7x^6\Rightarrow f'(x)=\frac{1+7x^6}{1+3(f(x))^2}$

Then,

$$f'(2)=\frac{1+7.64}{1+3(f(2))^2}=\frac{449}{1+3(f(2))^2}$$

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We also know $f(2) + (f(2))^3 = 2 + 2^7 = 130$, so $y = f(2)$ is a root to the polynomial $y^3 + y - 130 = 0$. This polynomial can be factored as$(y - 5)(y^2 + 5y + 26) = 0$. Since $y^2 + 5y + 26 = (y + 5/2)^2 + 79/4 > 0$

for all real y, the only real solution of $y^3 + y - 130 = 0$ is $y = 5$, so $f(2) = 5$. Hence,

$$\frac{a}{b} = f'(2) = \frac{449}{1+3.5^2} = \frac{449}{76}$$

Since 449 and 76 are relatively prime, the answer is $a + b = 449 + 76 = 525$

54) Volume $(V) = \frac{1}{3}A_1h_1 \Rightarrow h_1 = \frac{3V}{A_1}$

Same for h_2, h_3, h_4

So $(A_1 + A_2 + A_3 + A_4)(h_1 + h_2 + h_3 + h_4)$

$$= (A_1 + A_2 + A_3 + A_4)\left(\frac{3V}{A_1} + \frac{3V}{A_2} + \frac{3V}{A_3} + \frac{3V}{A_4}\right)$$

$$= 3V(A_1 + A_2 + A_3 + A_4)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)$$

A.M. $\geq$ H.M.

$$\frac{A_1 + A_2 + A_3 + A_4}{4} \geqslant \frac{4}{\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)}$$

$$\Rightarrow (A_1 + A_2 + A_3 + A_4)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right) \geqslant 16$$

Minimum value of $(A_1+A_2+A_3+A_4)\left(\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \frac{1}{A_4}\right)$

$$= 3V(16) = 48V = 48 \times 5 = 240 = \frac{240}{60} = 4$$

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